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Flexline wrap antenna for projectile

Abstract

A projectile circuitry assembly for use in projectiles comprising a chassis defining a generally cylindrical a main body portion and further defining an interior cavity for containing one or more projectile components and further defining an antenna aperture through the body portion to expose the interior cavity. In various embodiments the projectile circuitry assembly comprises a plurality of circuit boards and a wrap antenna, the plurality of circuit boards and wrap antenna interconnected via an integrated flex-line to define a single unitary device without the use of a connector, the wrap antenna comprising one or more antenna elements defined on a flexible antenna substrate layer, wherein the plurality of circuit boards are positioned in the interior cavity and the wrap antenna is threaded through the antenna aperture and wrapped circumferentially about an exterior of the cylindrical wall of the body portion.

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Background/Summary

RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 18/106,707, filed Feb. 7, 2023, which is a continuation of U.S. patent application Ser. No. 16/974,169, filed Oct. 29, 2020, issued as U.S. Pat. No. 11,581,632 on Feb. 14, 2023, which claims priority to U.S. Provisional Application No. 62/973,919, filed Nov. 1, 2019, the contents of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure relates to antenna systems for munitions, and more specifically, to side mounted or wrapped antenna systems for munitions.

BACKGROUND

(2) Extensive efforts have been directed toward guiding, steering, or configuring military grade projectiles for proximity sensing or seeking operations. Such projectiles greatly enhance target engagement and operational efficiencies compared to traditional projectiles. For example, in certain applications the ability to perform guided maneuvers and/or for proximity sensing may be necessary to provide a reasonable probability of engaging a target as delivery errors, environmental factors, or other issues are known to significantly degrade the effectiveness of traditional projectiles. This is particularly true when engaging moving targets, small targets, or targets that can take evasive action. In addition, capabilities can reduce collateral damage, conserve ammunition, reduces costs, minimize personnel time in engaging targets, among other benefits.

(3) Such projectiles have included barrel-fired and non-barrel-fired projectiles, boosted, and non-boosted projectiles, and spin-stabilized and fin-stabilized projectiles. In addition, such projectiles have included, low-caliber (50 caliber or less), medium-caliber (greater than 50 caliber to 75 mm), and large caliber projectiles (greater than 75 mm and generally used as artillery, rockets, and missiles).

(4) For example, large-caliber artillery and other projectiles, have been successfully guided—utilizing systems such as shown in U.S. Pat. No. 6,981,672, owned by the owner of the instant application. Artillery shells utilizing this type of design have been well received by the military. For example, see U.S. Pat. No. 7,412,930. These patents are incorporated herein by reference in their entireties for all purposes. Guided missiles have long been utilized for targeting aircraft and may be self-guided or remotely guided. See, for example U.S. Pat. No. 3,111,080, incorporated herein by reference in its entirety for all purposes. Such missiles are typically fin-stabilized rather than spin-stabilized, having internal propulsion systems and relying upon fins and radially-extending flaps or propulsion-directing members for altering flight path. In addition, guided missiles typically need to be launched or fired from launch tubes or brackets that are designed specific to the missile. Due to their internal propulsion systems, missiles are substantially more expensive than non-propelled projectiles.

(5) With respect to medium and small caliber projectiles, several solutions have been proposed utilizing movable aerodynamic surfaces for steering. For example, U.S. Pat. No. 6,422,507, incorporated herein by reference in its entirety for all purposes, discloses a greater than 50 caliber projectile that may be fired from a conventional barreled gun. This projectile utilizes a spoiler that extends and retracts from a rearwardly positioned despun portion out into the air stream. The despun portion is despun by a motor and batteries provide power to the bullet.

(6) Several solutions to guiding small-caliber projectiles, that is 50 caliber or less, have been proposed. These include firing the projectile without spinning the projectile and utilizing axially extending control fins for altering the flight. See, for example, U.S. Pat. No. 7,781,709, incorporated herein by reference in its entirety for all purposes. A notable disadvantage to such projectiles is that they cannot be fired from existing rifled barrels for conventional non-steerable projectiles and require internal batteries for operating the control circuitry and control fins which may affect the useful life of the projectile and provide a failure path. U.S. Pat. No. 5,788,178, incorporated herein by reference in its entirety for all purposes, also discloses a small-caliber bullet that is designed to be fired from a non-rifled barrel. Deployable flaps are utilized to control the flight path in the '178 device and the device requires a battery.

(7) U.S. Pat. No. 8,716,639 discloses small to medium caliber projectiles fired through a rifled barrel that use beveled surfaces or canards on a despun nose portion operated by a motor and battery for flight control. U.S. Pat. No. 4,537,371 discloses a projectile fired through a barreled

projectile that distributes air from the air stream through the projectile with valves to discharge the air laterally to change the flight path. These references are incorporated herein by reference in their entirety for all purposes. Additional prior guidance systems utilizing fins, wing-like projections, or canards have been proposed. See for example the following U.S. patents: U.S. Pat. Nos. 4,373,688; 4,438,893; 4,512,537; 4,568,039; 5,425,514; 6,314,886; 6,502,786; 7,849,800; 8,319,164. These patents are incorporated herein by reference in their entirety for all purposes.

(8) It is generally understood in the art that fuzing, sensing, communications, proximity, and other functions are generally required for such projectiles. For example, GPS, height-of-burst (HOB), sensing, seeking, proximity detection, and other functions add capabilities for control or to enhance projectile performance to engage a target. Further improvements are always welcome for these projectiles that enhance accuracy, allow miniaturization, increase range, provide cost savings, or improve reliability.

SUMMARY

(9) According to embodiments of the present disclosure, a flexible projectile circuitry assembly is disclosed for implementing a side-mounted or wrapped projectile antenna. In one or more embodiments, the flexible projectile circuitry assembly comprises a plurality of circuit boards and a wrap antenna, the plurality of circuit boards and wrap antenna interconnected via an integrated flex-line to define a single unitary device without the use of a connector, the wrap antenna comprising one or more antenna elements that are defined on a flexible antenna substrate layer. In various embodiments, the projectile circuitry assembly is configured for use with a projectile. In such embodiments, the projectile can comprise a nose portion with a forward tip, a body portion, a tail portion, and a chassis. In various embodiments the chassis extends from the tail portion to the nose portion and defines a generally cylindrical wall and an interior cavity within the projectile for containing one or more projectile components and further defining an antenna aperture through the cylindrical wall of the body portion exposing the interior cavity. In one or more embodiments, the plurality of circuit boards are positioned in the interior cavity and the wrap antenna is threaded through the antenna aperture and wrapped circumferentially about an exterior of the cylindrical wall of the body portion. In addition, in certain embodiments a sealing material, such as an epoxy, seals the antenna aperture.

(10) A significant difficulty in the design of guided or smart projectiles is the implementation of side mounted or wrapped antennas onto projectiles. Such designs offer various advantages and compatibility with numerous types of projectile designs, however, with increasing environmental stresses such as high temperatures, g-force stresses, rough handling, and the like, it is difficult to successfully mount an antenna on the side of a projectile and keep the antenna in operation after firing.

(11) As such, embodiments of the present disclosure provide an advantageous solution as compared to other implementations of side-mounted projectile antennas. For example, referring to FIG. 1, a cross-sectional front view of a projectile **100** a previously utilized implementation of a side-mounted projectile antenna. As shown, the projectile **100** has a generally cylindrical shape that includes a projectile sidewall **104**. The sidewall **104** defines a projectile exterior and an interior space **108** within the projectile **100** for containing various projectile circuitry, payload, fuzing, and/or other components. Depicted in FIG. 1, the projectile **100** includes a projectile circuitry assembly **112** having an exterior wrap antenna portion **116** that is coupled with an interior antenna circuitry portion **120** via an antenna connector **124** mounted in the projectile sidewall **104**. The interior antenna circuitry portion **120** includes one or more interconnected boards **122**, each including various circuitry, and a connector **126** coupled with a flex-line material **128** of the one or more interconnected boards. Similarly, the wrap antenna portion **116** is comprised of one or more antenna elements defined on a flexible substrate **132**. The substrate **132** is wrapped around the exterior of the projectile sidewall **104** and includes another connector **134** coupled with a flex-line material of the substrate **132**. Each connector **126**, **134** is soldered to be coupled with their

respective flex-line material—generally requiring at least two solder joints each. In addition, to assemble the antenna portion **116** with the antenna circuitry, the connectors **126**, **134** are additionally soldered to the antenna connector **124** mounted in the sidewall **104**—generally requiring at least two solder joints each. Finally, the antenna connector **124** itself is soldered to the projectile sidewall **104** to mount the antenna connector **124** in place.

(12) Furthermore, during assembly of the projectile **100**, some of these solder joints or cable connections must be performed while the boards are located within the projectile **100**. This requires precise manual soldering which is very difficult and expensive. In addition, the higher the frequency of the antenna, the more expensive this solution gets. For example, at higher frequencies, the connectors can get extremely expensive, as well as it is difficult to find connectors that are small enough to fit within a 20 mm or 30 mm projectile. In particular, anything beyond 6 GHz becomes very difficult to produce and solder. Along with this of course, you can have mating problems between connectors, repeatability issues, problems with the connectors or cables themselves, too much strain on the cables so that they get damaged, and the extra space it all takes up. Finally, these problems are compounded where the projectile includes more than one externally wrapped antenna.

(13) As such, embodiments of the disclosure provide for an improved design for antenna control circuitry and antenna subsystems that eliminate the need for complicated soldering and is scalable for use in 20 mm or smaller projectiles. Also, this invention can also easily be adapted to much higher frequencies (up to and beyond 40 GHz without difficulty) with the precision of etching tolerances on circuit boards/flex-lines. The above summary is not intended to describe each illustrated embodiment or every implementation of the present disclosure.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) The drawings included in the present application are incorporated into, and form part of, the specification. They illustrate embodiments of the present disclosure and, along with the description, serve to explain the principles of the disclosure. The drawings are only illustrative of certain embodiments and do not limit the disclosure.

(2) FIG. **1** depicts a cross-sectional front view of a projectile that includes a previously utilized implementation of a side-mounted projectile antenna.

(3) FIGS. **2A-2E** depict perspective views, plan views, and partial cross-sectional views of a projectile, according to one or more embodiments of the disclosure.

(4) FIGS. **3A-3C** depict perspective views of a projectile circuitry assembly in an unfolded and in various compact states for insertion into a projectile chassis, according to one or more embodiments of the disclosure.

(5) FIGS. **4A & 4B** depict side views of a flex-board in stages of manufacture into a projectile circuitry assembly, according to one or more embodiments of the disclosure.

(6) FIGS. **5A & 5B** depict side views of a flex-board in stages of manufacture into a projectile circuitry assembly, according to one or more embodiments of the disclosure.

(7) FIGS. **6A-6D** depict stages of assembly of a projectile circuitry assembly with a projectile chassis, according to one or more embodiments of the disclosure.

(8) FIG. **7** depicts assembly of a projectile circuitry assembly with a projectile chassis, according to one or more embodiments of the disclosure.

(9) FIGS. **8A & 8B** depict an assembled projectile chassis and projectile circuitry assembly, according to one or more embodiments of the disclosure.

(10) While the embodiments of the disclosure are amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will

be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

(11) Referring to FIGS. 2A-2E, perspective views, plan views, and partial cross-sectional views of a projectile is depicted, according to one or more embodiments of the disclosure. In various embodiments, the projectile **200** includes a main body portion **204**, a tail portion **208**, and a nose portion **212**. A projectile sidewall or projectile chassis **214** defines at least the main body portion **204** and can additionally define the tail portion **208** and/or nose portion **212**.

(12) Described further below, the projectile **200** additionally includes a projectile circuitry assembly **216**. In various embodiments the projectile circuitry assembly **216** is a collection of components configured to perform one or more various functions for the projectile **200** including, but not limited to, communication, sensing, proximity detection, and fuzing. Described further below, in various embodiments the projectile circuitry assembly **216** includes an antenna circuitry portion **218** and a wrap antenna portion **220**. In one or more embodiments the antenna circuitry portion **218** includes various antenna circuitry, such as one or more transmitters and receivers, that can be used to transmit/receive signals at respective frequencies for the projectile circuitry assembly **216**. While the antenna circuitry portion **218** includes various circuitry for functionality of communication and antenna operation, in various embodiments, the antenna circuitry portion **218** can additionally or optionally include various other kinds of projectile control circuitry, such as sensing components, processing components, fuzing, or other electronic components.

(13) In various embodiments, the circuitry is distributed among one or more interconnected circuit boards **222**. Described further below, in various embodiments the one or more boards **222** are interconnected via an integrated flex-line **224**. As used herein, flex-line **224** refers to a signal conducting portion of the assembly that is configured to connect the various portions of the assembly and is flexibly bendable to allow for folding or positioning of the various elements of the assembly into a compact shape for insertion into a projectile chassis.

(14) In such embodiments, the boards **222** and flex-line **224** are formed from a single piece of material, including various layers that are selectively etched or removed to define the board portions of the circuitry portion **218** and the flex-line interconnects. Similarly, and described further below, the wrap antenna portion **220** is comprised of one or more antenna elements that are defined on an antenna substrate layer. In some embodiments, the antenna can be deposited by 3D printing or other metal deposition methods

(15) In various embodiments, the main body portion **204** has a cylindrical shape or a generally cylindrical shape defined by the chassis **214** and has an exterior surface **232**, a forward portion **236** and a rearward portion **240**. In some embodiments, chassis **214** defines one or more tapered portions that converge in a direction along a central axis **244**. For example, in some embodiments a first portion, such as the forward portion **236** converges in a forward direction, along central axis **244**, towards the nose portion **212**. In some embodiments, a second portion, such as the rearward portion **240** could converge in a rearward direction towards the tail portion **108**. The chassis **214** is, in some embodiments, machined or formed from a single block of metal. However, in certain embodiments the chassis **214** can include a plurality of pieces that are fastened, screwed, or otherwise assembled together.

(16) In various embodiments the chassis **214** defines an antenna recess **250** in the main body portion **204**. Described further below, in one or more embodiments the wrap antenna portion **220** of the projectile circuitry assembly **216** is placed or mounted within the antenna recess **250**. Depicted at least in FIG. 2C, the antenna portion **220** at least partially fills the recess **250** and leaves a gap or an unfilled portion that is defined between the antenna portion **220** and the exterior **232** of the main body portion **204**. In such embodiments, the gap or unfilled portion of the antenna recess **250** defines a buffer space. In various embodiments, when the projectile **200** is fired, the buffer space

provides a space between the barrel of a gun and the antenna portion **220** that serves to protect the antenna from damage during projectile firing. In addition, in various embodiments a protective cover **252** can be wrapped over or included with the antenna portion **220** to further protect the antenna portion **220** from damage during projectile firing. In certain embodiments, the protective cover **252** can be deposited onto the antenna **220** by common deposition methods. Depicted in FIGS. 2A and 2B, the protective cover is shown while in FIGS. 2C and 2D the protective cover is removed to reveal the antenna portion **220**.

(17) In one or more embodiments the chassis **214** defines a cavity **256** within the projectile **200** for supporting/containing one or more projectile components. As an example, antenna circuitry portion **218** is positioned is depicted in FIG. 2D positioned in cavity **256** of the main body portion **104**. As described above, in various embodiments, the antenna circuitry portion **218** can include a variety of projectile control circuitry such as communication componentry, sensing components, processing components, or other components of the projectile **200**. In such embodiments, any or all of these components could additionally be positioned in the cavity **256**.

(18) In one or more embodiments, the chassis **214** defines an antenna aperture **260** or opening that exposes the interior cavity **256** of the projectile **200** from the antenna recess **250**. Described further below, the antenna aperture **260** allows for projectile circuitry assembly **216** to extend between the interior and exterior of the projectile **200**, with the wrap antenna portion **220** of the projectile circuitry assembly **216** circumferentially wrapped about the projectile **200** exterior in the antenna recess **250** while also extending through the aperture **260** to connect with the interior antenna circuitry portion **218**. Described further below, in one or more embodiments, a sealing material, such as an epoxy filler **262** and/or solder is used to seal the aperture **260** in assembly of the antenna **218** with the projectile chassis **214**.

(19) The nose portion **212** is a forward-facing structure and has a tapered or a converging shape. As such, the nose portion **212** extends from the forward portion **236** of the main body portion **204**, forwardly, in a first direction, along central axis **244** to a forward tip portion **268**. In various embodiments, nose portion **212** has an exterior surface **270** and may be conical or have a curved taper from the forward portion **236** of the main body portion **204** to the forward tip portion **268**. In various embodiments, the nose portion **212** is removable from the chassis **214** to reveal or expose an opening **272** into the interior cavity **256** of the projectile **200**. Described further below, the opening **272** allows for initial insertion of the projectile circuitry assembly **218**, such as during projectile assembly.

(20) In one or more embodiments, the projectile **200** can include one or more flight control portions for directing or otherwise altering the trajectory of the projectile during flight. For example, in certain embodiments, the chassis could define a control support that supports a rotatable collar with one or more aerodynamic features for despinning and/or flight control. In such embodiments, alternator components or other power generation components could additionally be implemented to utilize the spinning collar for power generation/power supply for various electronic projectile components. Examples of power supply and projectile control mechanisms can be found in the various patent applications and/or publications incorporated by reference below.

(21) Referring to FIGS. 3A-3C, various perspective views of a projectile circuitry assembly **300** are depicted, according to one or more embodiments. Projectile circuitry assembly **300** is substantially similar to the projectile circuitry assembly **200** described above with references to FIGS. 2A-2E, as such, in various embodiments, the projectile circuitry assembly **300** is a collection of components configured to perform one or more various functions for a projectile including, but not limited to, communication, sensing, proximity detection, and fuzing.

(22) As described above, in various embodiments, the projectile circuitry assembly **300** includes an antenna circuitry portion **304** and a wrap antenna portion **308**. In one or more embodiments the antenna circuitry portion **304** includes various antenna circuitry, such as one or more transmitters and receivers, processors, memory, fuzing or other components for executing various projectile

functions. In various embodiments, the circuitry is distributed among one or more of a plurality of interconnected circuit boards **312** that are interconnected via an integrated flex-line **316**. In such embodiments, the boards **312** and flex-line **316** are formed from a single piece of material, including various layers that are selectively etched or removed to define the boards **312** and the flex-line interconnects **316**. Similarly, the wrap antenna portion **308** is comprised of one or more antenna elements that are defined on an antenna substrate layer.

(23) In contrast with the previously utilized implementation of a side-mounted projectile antenna described above with respect to FIG. 1, the projectile circuitry assembly **300** is additionally integrated with the circuitry portion **304** without the use of a separate connector or other device. As such, the projectile circuitry assembly **300** is formed from a unitary piece of material that includes both the circuitry portion **304** and respective antenna portion **308** as a single unitary piece including various layers that are selectively etched or removed to define the antenna element **308** and flex-line interconnect **316**. In various embodiments, the integral nature of the assembly **300** allows the device to be easily folded or configured into a compact state, to allow for easy assembly with a projectile chassis without requiring any soldering connections between the various components of the projectile circuitry assembly **300**. For example, referring to FIG. 3B, the board is depicted in a first compact state with the flex-line interconnects folded to arrange boards **312** in a stack and antenna portion **308** wrapped around the boards **312** to present a compact footprint for insertion in and assembly with a projectile chassis. Similarly, referring to FIG. 3C, the board is depicted in another compact state with the flex-line interconnects folded to arrange boards **312** in a stack, but without the antenna portion **308** wrapped about the boards. Instead, described further below, the antenna portion **308** is arranged for insertion into a slotted antenna aperture for wrapping about the projectile chassis.

(24) Referring to FIGS. 4A and 4B, side views of a flex-board are depicted in stages of manufacture into a projectile circuitry assembly, according to one or more embodiments. Referring specifically to FIG. 4A a layered flex-board **400A** is depicted prior to etching or removal of material to define a projectile circuitry assembly. In one or more embodiments, the board **400A** includes a plurality of layers of various types of material. Depicted in FIG. 4A, the flex-board **400A** includes nine layers that define at least three distinct board portions including a circuit board portion **404**, a flex-line portion **408**, and an antenna portion **412**. In such embodiments, the board portion **404** can be defined by at least a board ground layer **414**, a board substrate layer **416**, and a first transition layer **418**. In one or more embodiments, the board ground layer **414** functions as a ground for various attached components and may, in various embodiments, include components attached to the layer **414**. In one or more embodiments, the board substrate layer **416** functions as a substrate for board components. In various embodiments, the first transition layer **418** is positioned between the board portion **404** and flex-line portion **408** and functions both as a board signal layer and as a ground layer for the bottom of the flex-line portion **408**. In such embodiments, the first transition layer **418** can connect various components for signal transfer.

(25) In various embodiments, the flex-line portion **408** can be defined by at least a pair of flex-line substrate layers **420**, a flex-line signal layer **422**, and the first transition layer **418** and a second transition layer **424**. In such embodiments, the flex-line substrate layers **420** function as top and bottom substrates for the flex-line while the signal layer **422** transmits signal lines along the flex-line portion **408**. As described above, the first transition layer **418** functions as a ground layer for the bottom of the flex-line portion **408**. Similarly, the second transition layer **424** is positioned between the antenna portion **412** and flex-line portion **408**. In various embodiments, the second transition layer **424** functions as both an antenna ground layer for the antenna portion **412** or as a flex-line ground layer for the top-side of the flex-line portion **408**. In various embodiments, the antenna portion **412** is defined by at least an antenna layer **430**, an antenna substrate **432**, and the second transition layer **424**. In various embodiments, the antenna layer **430** is configured as an antenna, such as a patch antenna, and is configured to transmit/receive an antenna signal. In one or

more embodiments, the antenna substrate layer **432** functions as the antenna substrate, and as described above, the second transition layer **424** would function as well as the ground of the antenna portion **412**.

(26) Depicted in FIG. **4B**, the flex-line board **400A** can be machined, etched, cutout, or otherwise formed into a board **400B** for the appropriate requirements of the projectile circuitry assembly. Depicted in FIG. **4B**, the board **400B** now includes a circuit board **434** defined on the far left, a flex-line **436** on either side of the circuit board **434** and connecting the circuit board **434** to an antenna **438** positioned on the right. In this configuration, one or more additional circuit boards **434** could additionally be defined to the left with the flex-line **436** connecting each of the boards, various circuitry on the boards, and the antenna portion **438** together as a unitary device.

(27) Referring to FIGS. **5A** and **5B**, side views of a flex-board are depicted in stages of manufacture into a projectile circuitry assembly, according to one or more embodiments. Referring specifically to FIG. **5A** a layered flex-board **500A** is depicted prior to etching or removal of material to define a projectile circuitry assembly.

(28) In one or more embodiments, the board **500A** includes a plurality of layers of various types of material. Depicted in FIG. **5A**, the flex-board **500A** includes seven layers that define at least two board portions including a circuit board portion **504** and a signal portion **508**. In such embodiments, the board portion **504** can be defined by at least a board ground layer **514**, a board substrate layer **516**, and a first transition layer **518**.

(29) In one or more embodiments, the board ground layer **514** functions as a ground for various attached components and may, in various embodiments, include components attached to the layer **514**. In one or more embodiments, the board substrate layer **516** functions as a substrate for board components. In various embodiments, the first transition layer **518** is positioned between the board portion **504** and signal portion **508** and functions both as a board signal layer and as a ground layer for the bottom of the flex-line portion **508**. In such embodiments, the first transition layer **518** can connect various components for signal transfer.

(30) In various embodiments, the signal portion **508** can be defined by a pair of substrate layers including a first substrate layer **520**, a second substrate layer **522**, a signal layer **524**, a ground layer **526** and the first transition layer **518**. In such embodiments, the signal portion **508**, at least based on the etching or machining performed on the flex-line board **500A** is configurable to function as either a flex-line **530** or antenna **534**. For example, depicted in FIG. **5B**, in various embodiments, when configured as a flex-line **530**, the first and second substrate layers **520**, **522** function as top and bottom substrates while the signal layer **524** functions to transmit signal lines along the flex-line **530**. When configured as an antenna **534**, in various embodiments the first transition layer **518** functions as a ground layer for the bottom of the antenna **534**, the first substrate layer **520** functions as an antenna substrate, and the flex-line signal/antenna layer **524** functions as an antenna, such as a patch antenna, and is configured to transmit/receive an antenna signal.

(31) Depicted in FIG. **5B**, the flex-line board **500A** can be machined, etched, cutout, or otherwise formed into a board **500B** for the appropriate requirements of the projectile circuitry assembly. Depicted in FIG. **5B**, the board **500B** now includes a circuit board **528** defined on the far left, a flex-line **530** on either side of the circuit board **528** and connecting the circuit board **528** to an antenna **534** positioned on the right. In this configuration, one or more additional circuit boards **528** could additionally be defined to the left with the flex-line **530** connecting each of the boards, various circuitry on the boards, and the antenna portion **534** together as a unitary device.

(32) Referring briefly to FIGS. **4A**, **4B**, **5A**, and **5B**, these are just two potential flex-line board configurations of many. In various embodiments, the flex-line boards could include additional or fewer layers, for example based on the needs of the user. Furthermore, depending on what materials are used on the boards, they both could be used for high frequency (more than 15 GHz) or low frequency (less than 15 GHz). In certain embodiments, the flex-line boards will generally be made of a less expensive RF material, such as FR4, which does not perform as well when frequencies get

very high. However, this FR4 material could just as easily be replaced with better RF performing substrates, as known in the art, based on the requirements of the board.

(33) Referring to FIG. 6A-6D, stages of assembly of a projectile circuitry assembly **300** with a projectile chassis **214** is depicted, according to one or more embodiments. In one or more embodiments, the projectile circuitry assembly **300** is configured in a compact state with the boards **312** arranged together in a stack and antenna portion **308** wrapped around the boards **312** to present a compact footprint for insertion and assembly with a projectile chassis **214**. As described above, in one or more embodiments the chassis **214** defines a cavity **256** within the projectile **200** for supporting one or more projectile components. In various embodiments, an opening **272** into the interior cavity **256** is defined at a forward portion **236** of the chassis **214**. Referring specifically to FIG. 6A, in a first stage of assembly the opening **272** allows for initial insertion of the folded projectile circuitry assembly **300** into the interior of the projectile **200**. As described above, in one or more embodiments, the chassis **214** defines an antenna aperture **260** or opening that exposes the interior cavity **256** of the projectile **200** from the antenna recess **256**. In such embodiments the antenna aperture **260** allows for projectile circuitry assembly **300** to extend between the interior and exterior of the projectile **200**, with the wrap antenna portion **308** of the projectile circuitry assembly **300** circumferentially wrapped about the projectile **200** exterior in the antenna recess **250** while also extending through the aperture **260** to connect with the interior antenna circuitry.

(34) Referring specifically to FIG. 6B, a second stage of assembly is depicted, according to various embodiments. As shown, once inserted, the projectile circuitry assembly **300**, with antenna portion **308** wrapped around the boards **312**, is rotated to thread the antenna portion **308** of the assembly **300** through the aperture **260**. As seen in FIG. 6C, the projectile circuitry assembly **300** is rotated to thread the antenna portion **308** through the aperture **260** and wrap the projectile circuitry assembly **308** about the exterior of the chassis **214**.

(35) Referring specifically to FIG. 6D, in one or more embodiments, a sealing material, such as an epoxy filler **262** and/or solder **604** is used to seal or otherwise fill the aperture **260** in assembly of the antenna **218** with the projectile chassis **214**. In various embodiments, the solder **604** encloses the one or more circuit boards in the projectile interior and isolates the one or more circuit boards from exterior electromagnetic fields.

(36) Referring to FIG. 7, stages of assembly of a projectile circuitry assembly **300** with a projectile chassis **714** is depicted, according to one or more embodiments. In one or more embodiments, the projectile circuitry assembly **300** is configured in a compact state with the boards **312** arranged together in a stack and antenna portion **308** wrapped around the boards **312** to present a compact footprint for insertion and assembly with a projectile chassis **714**. As described above, in one or more embodiments the chassis **714** defines a cavity **256** within the projectile **200** for supporting one or more projectile components. In various embodiments, an opening **272** into the interior cavity **256** is defined at a forward portion **236** of the chassis **714**.

(37) In a first stage of assembly the opening **272** allows for initial insertion of the folded projectile circuitry assembly **300** into the interior of the projectile **200**. As described above, in one or more embodiments, the chassis **714** defines an antenna aperture **760** or opening that exposes the interior cavity **256** of the projectile **200** from the antenna recess **256**. Depicted in FIG. 7, the antenna aperture **760** is slotted, defining a forward opening **768** that allows for insertion of the antenna portion **308** without requiring the wrapping and rotation of the antenna assembly, described above. In such embodiments the antenna aperture **760** allows for projectile circuitry assembly **300** to extend between the interior and exterior of the projectile **200**, as described above.

(38) The result of assembly is depicted in FIGS. 8A & 8B, as described above, in various embodiments, once the antenna portion **308** is inserted and threaded through and secured in place, a protective layer **252** can be added to protect the antenna from damage during projectile firing. In various embodiments a protective cover **252** can be wrapped over or included with the antenna portion **220** to further protect the antenna portion **220** from damage during projectile firing

(39) In addition to the above disclosure, the disclosure of U.S. Pat. No. 6,981,672, which is owned by the owner of this application, is fully incorporated by reference herein for all purposes. Also incorporated herein for all purposes, in their entireties are: U.S. Pat. Nos. 6,422,507; 7,412,930; 7,431,237; 6,345,785; 8,916,810; 6,653,972; 7,631,833; 7,947,936; 8,063,347; 9,709,372; 9,683,814; 8,552,349; 8,757,064; 8,508,404; 7,849,797; 7,548,202; 7,098,841; 6,834,591; 6,389,974; 6,204,801; 5,734,389; 5,696,347; 6,502,786; 6,666,402; 6,693,592; 7,681,504; 8,319,163; 8,324,542; 8,674,277; 8,950,335; 9,303,964; 9,360,286; 9,557,405; 9,587,923; 10,054,404; 2006/0061949; 2018/0245895; and 2019/0041527.

(40) The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A projectile having a nose portion with a forward tip, a body portion, a tail portion, and a central axis, the projectile comprising: a chassis extending from the tail portion to the nose portion, the chassis defining a generally cylindrical wall of the body portion and further defining an interior cavity within the projectile and further defining an antenna aperture through the cylindrical wall of the body portion; a projectile circuitry assembly comprising one or more circuit boards and a wrap antenna, the one or more circuit boards and the wrap antenna interconnected via an integrated flex-line, the wrap antenna comprising one or more antenna elements that are defined on a flexible antenna substrate layer, wherein the one or more circuit boards are positioned in the interior cavity and at least a portion of the wrap antenna projects through the antenna aperture and is wrapped circumferentially about an exterior of the cylindrical wall of the body portion; and a sealing material sealing the antenna aperture.
2. The projectile of claim 1, wherein the projectile circuitry assembly comprises: a board ground layer, a board substrate layer, a first transition layer, a first flex-line substrate layer, a flex-line signal layer, a second flex-line substrate layer, a second transition layer, an antenna substrate layer, and an antenna signal layer.
3. The projectile of claim 2, wherein the projectile circuitry assembly is a flex-line board comprising: a circuit board portion, a flex-line portion, and an antenna portion, the circuit board portion defined by the board ground layer, the board substrate layer, and the first transition layer, the flex-line portion defined by the first transition layer, the first and second substrate layers, the flex-line signal layer, and the second transition layer, and the antenna portion defined by the second transition layer, the antenna substrate layer, and the antenna signal layer.
4. The projectile of claim 1, wherein the projectile circuitry assembly is a flex-line board comprising: a board ground layer, a board substrate layer, a first transition layer, a first substrate layer, a second substrate layer, a signal layer, and a ground layer.
5. The projectile of claim 4, wherein the projectile circuitry assembly comprises a circuit board portion and a signal portion, the circuit board portion defined by at least a board ground layer, a board substrate layer, and the first transition layer, and the signal portion defined by the first substrate layer, second substrate layer, the signal layer, the ground layer, and the first transition layer.
6. The projectile of claim 5, wherein the signal portion, is configurable to function as either a flex-

line or an antenna.

7. The projectile of claim 1, wherein the generally cylindrical wall of the body portion defines an antenna recess in the main body portion; and wherein the antenna portion is mounted within the antenna recess such that the antenna portion partially fills the recess and leaves an unfilled portion between an exterior of the antenna recess and an exterior of the cylindrical wall.

8. The projectile of claim 1, wherein the wrap antenna further comprises a protective cover.

9. The projectile of claim 1, wherein the one or more circuit boards define an antenna circuitry portion, wherein the antenna circuitry portion includes one or more projectile control circuitry comprising sensing components, processing components, baseband processing components, and fuzing.

10. The projectile of claim 1, wherein the sealing material comprises an epoxy.

11. The projectile of claim 1, wherein the sealing material comprises a solder bead.

12. The projectile of claim 11, wherein the solder bead encloses the one or more circuit boards in the projectile interior and isolates the one or more circuit boards from exterior electromagnetic fields.

13. A method of assembling a projectile and a projectile circuitry assembly, the projectile comprising a chassis extending from a tail portion to a nose portion, the chassis defining a generally cylindrical wall of a main body portion and further defining an interior cavity within the projectile for containing one or more projectile components and further defining an antenna aperture slot extending axially through the cylindrical wall of the body portion exposing the interior cavity, and the projectile circuitry assembly comprising one or more circuit boards and a wrap antenna, the one or more circuit boards and the wrap antenna interconnected via an integrated flex-line to define a unitary device, the wrap antenna comprising one or more antenna elements that are defined on a flexible antenna substrate layer, wherein the method comprises: configuring the projectile circuitry assembly into a compact state such that the plurality of circuit boards are insertable into the interior cavity; disposing the antenna to extend through the antenna aperture slot; wrapping the antenna of the projectile circuitry assembly about the exterior of the chassis; and sealing the antenna aperture slot with a sealing material.

14. The method of claim 13, wherein in the compact state the integrated flex-line is folded to arrange the one or more circuit boards together in a stack with the antenna portion wrapped around the one or more circuit boards.

15. The method of claim 13, wherein the generally cylindrical wall of the body portion defines an antenna recess in the main body portion; and wherein the antenna is wrapped within the antenna recess such that the antenna portion partially fills the recess and leaves an unfilled portion between an exterior of the antenna recess and an exterior of the cylindrical wall.

16. The method of claim 13, wherein the sealing material comprises an epoxy.

17. The method of claim 13, wherein the sealing material comprises a solder bead.

18. The method of claim 17, wherein the solder bead encloses the one or more circuit boards in the projectile interior and isolates the one or more circuit boards from exterior electromagnetic fields.

19. The method of claim 13, wherein the antenna aperture slot defines a forward opening at the opening into the interior cavity.

20. The method of claim 19, wherein the wrap antenna is inserted at the forward opening of the antenna aperture slot.
